Commissioner of Patents
PO Box 200
Woden ACT 2606

Laminar IP
PO Box 1188
North Sydney NSW 2059
Telephone: 02 8379 1281
Email: mail@laminarip.com

11 September 2026

Our Ref: P120867AU
Contact: Jon Wright

Dear Commissioner,

Australian Patent Application No. 2026201831
Letsdex Pty Ltd
CROSS-CHAIN STATE VERIFICATION METHOD, APPARATUS, ELECTRONIC DEVICE, AND STORAGE MEDIUM FOR BLOCKCHAIN

Response to Examination Report
The Examination Report dated 1 May 2026 is acknowledged.

The Applicant seeks leave of the Commissioner under Section 104 to amend the application in accordance with the attached Statement of Proposed Amendments.

Without prejudice, the Applicant proposes to amend the claims in order to advance prosecution.

Specifically, former Claims 4-5 and 10 are proposed to be amended.

The patent specification and claims are also proposed to be amended to align with local practice.

The Applicant respectfully submits that no new subject matter is introduced by way of these proposed amendments.

We comment as follows in regard to the objections taken in that Report.

Item 1 – Section 40 (clarity)

The Examiner asserts that Claims 4-5 and 9-10 lack clarity.

Specifically, the Examiner considers that Claim 4 is unclear because it recites performing validity verification "*on a public user key*", whereas signature verification would ordinarily be performed *using* a public key, and the description appears to support that interpretation.

Claim 5 is said to be unclear because the phrase "*collateral assets corresponding to the user*" does not make it clear whether the collateral belongs to the user who submitted the state assertion or to a validator whose conduct is being penalised.

Claim 5 is also said to lack clarity because it refers to "*the state verification result*" when dependent on Claim 2, even though that term is only introduced in Claim 4, creating an antecedent basis issue.

Finally, Claims 9 and 10 are considered unclear because Claim 9 defines a "*target threshold*" as potentially being several different kinds of threshold (such as a validator quantity threshold, consensus ratio threshold, or verification score threshold), but Claim 10 allegedly, does not make clear whether the stated proportional relationships apply to all of those threshold types or only some of them.

Without acquiescing to the Examiner's position, the Applicant proposes to amend the claims in order to advance prosecution.

Specifically, former Claim 4 is proposed to be amended to recite:

> "*…performing signature validity verification <u>using</u> ~~on~~ a user public key…*"

Former Claim 5 is amended to depend on Claim 4, and further amended to recite:

> "*…confiscating collateral assets corresponding to the user <u>who submitted the state assertion</u>…*"

While former Claim 10 is proposed to be amended to recite:

> "*…<u>wherein</u> the ~~target threshold is~~ <u>validator quantity threshold, the consensus ratio threshold and the verification score threshold, are each</u>:*
>
> - *directly proportional to the economic value of the business operation;*
> - *~~the target threshold is~~ directly proportional to the risk level of the business operation;*
> - *~~the target threshold is~~ inversely proportional to the user collateral asset ratio; and*
> - *~~the target threshold is~~ inversely proportional to the time urgency of source chain state changes*."

The Applicant respectfully submits that by virtue of these proposed amendments, the lack of clarity objection is rendered moot and should be withdrawn.

Item 2 – Inventive Step

The Examiner asserts that:

- *Claims 1-2 and 17-18 lack an inventive step in view of the combined disclosures of D1: CN 113806699 A (NAT UNIV DEFENSE TECHNOLOGY PLA) and D2: US 2020/0137064 A1 (EMC IP HOLDING CO LLC); and*

- *Appended Claims 3-16 and 19-20 add only features that are either disclosed in D1 and/or D2 or are considered to form part of the common general knowledge in the art, or mere design choices.*

The Applicant respectfully submits that while the Examiner concedes that D1 does not disclose storing the digitally signed verifiable credential in a distributed storage system, validators accessing that storage using a credential identifier, or the claimed verifiable credential structure, the present invention is distinguished from both D1 and D2 at a more fundamental level.

Present Application

The presently claimed invention is directed to a specific framework for **cross-chain state verification**, of which state consistency verification against source-chain data is the defining step, whereas D1 and D2 are directed principally to identity and credential verification.

For instance, the core of the invention defined in Claims 1 and 2 is the use of a verifiable credential containing a **state assertion regarding verifiable data on a source chain**, together with a content hash and a digital signature, to enable a destination-chain system to determine whether a factual state asserted to exist on another blockchain is true.

Claims 1 and 2 expressly require that the state assertion be an assertion regarding verifiable data on the source chain and that validators perform verification based on that credential.

The specification repeatedly explains that the invention concerns assertions regarding objectively verifiable on-chain facts such as asset ownership, balances, smart contract states, transaction histories, governance rights and other blockchain state data. Verification is performed by comparing the asserted state against the actual state existing on the source chain.

In essence, the presently claimed invention combines three (3) central elements that hold together as a single logical chain:

(i) The nature of the credential content.

The claimed Verifiable Credentials (hereinafter referred to as "VC") carries a state assertion about verifiable data on a source chain — a statement of on-chain fact (e.g. ownership of an asset, an account balance), not an identity attribute.

(ii) Verifying a fact requires a verification step that verifying an identity does not.

The claimed method's state consistency verification — comparing the state assertion against the actual verifiable data on the source chain.

(iii) The VC structure — embedding a content hash of the state assertion, signed with the user's private key — is what makes independent, trustless verification of that fact possible.

> *Any validator can retrieve the VC from distributed storage via its credential identifier, independently re-compute the hash of the state assertion, verify the signature against the DID-resolved public key, and independently check the assertion against the source chain — without any relay, intermediary, or trusted storage layer*

The Applicant respectfully submits that neither D1 nor D2 discloses or fairly suggests either of these elements.

CN 113806699 A (D1)

For instance, D1 is directed to a decentralised identifier and verifiable credential framework in which claims and credentials relate to identity, credentials and descriptive characteristics of an entity. Specifically, D1 concerns cross-blockchain verification using DIDs and verifiable credentials and a consensus process for acceptance or rejection of claims.

That is, D1 is a cross-chain identity verification system:

> its Claim object's ClaimContent records "*description characteristics*" of an entity — who an entity is (D1 [0014]–[0017]),

where D1's signers and verifiers "*verify*" a claim and record a binary accept/reject stance (the Stand field).

However, D1 never defines what is verified or how; nothing in D1 involves comparison against source-chain state.

US 2020/0137064 A1 (D2)

D2 also concerns decentralised identity management, certificate verification, public-key verification and authorization of requests within an identity framework.

The Examiner relies on D2 for teaching decentralised storage and public/private key verification, but D2 remains directed to identity management rather than verification of factual blockchain state.

D2's verification is signature + certificate + permission checking in an identity/permission context (D2 [0079], [0083] [0095]).

In other words, D1 is an identity attribute, while D2 is a single-chain identity management system with no cross-chain dimension at all.

It is evident therefore, that neither D1 nor D2 teaches a credential designed to be independently fact-checked in this way, because neither addresses the problem of attesting to on-chain facts across chains.

That is, neither D1 nor D2 addresses the problem identified in the present specification, namely improving the security of **cross-chain state verification** by enabling independent verification of asserted source-chain facts.

Neither D1 nor D2 teaches or suggests state consistency verification

The distinction is not merely the content of the credential.

Verification of a factual assertion requires a verification process fundamentally different from identity verification.

The present specification teaches that validators verify whether a state assertion is truthful by querying the source chain, determining the actual state of the relevant verifiable data, and comparing that actual state with the state assertion contained in the credential.

If the two are inconsistent, the assertion is determined to be fraudulent.

This concept is reflected expressly in Claim 4, which defines a state consistency verification step performed on the state assertion and the underlying verifiable data.

Neither D1 nor D2 teaches or suggests such a verification process.

As described in the Examination Report, D1 discloses retrieval of claims and publication of acceptance or rejection results through consensus, but does not disclose comparison of a credential-contained state assertion against actual source-chain state data.

Likewise, D2 performs certificate validation, permission checks and signature verification.

The Examiner identifies D2 only as teaching distributed storage together with public/private key verification.

D2 does not disclose querying a source blockchain and comparing actual blockchain state against an asserted state contained in a credential.

The feature which characterises the invention is therefore not merely storage of credentials in a distributed storage system, but the creation of a credential that enables trust-less verification of cross-chain factual assertions through subsequent validation against the true state of the source chain.

Neither cited document contains any teaching or suggestion of that concept.

The Examiner's combination does not lead to the claimed invention

Even accepting, for argument's sake, that the skilled person would have adopted D2's distributed-storage architecture within D1 (which the Applicant does not concede), the result would merely be an identity or credential verification framework in which credentials are stored off-chain.

It would not produce a system in which:

- *a credential contains a state assertion regarding verifiable data on a source chain;*
- *the state assertion is independently verifiable against actual source-chain data; and*
- *validators determine whether that asserted state is true by reference to the source chain itself.*

The Examiner's rationale therefore only addresses the storage location of credential data and the use of cryptographic verification techniques.

It does not explain why the skilled person would modify D1's identity-focused claim verification framework so that it becomes a mechanism for verifying objective source-chain facts.

The proposed combination relies on hindsight

The Applicant further submits that the proposed combination is based on impermissible hindsight.

D1 does not identify storage of verifiable credentials as a problem, nor does D1 suggest relocating credential data to an off-chain distributed storage system.

The motivation identified by the Examiner is taken from D2, which addresses storage and scalability issues in a different context, namely decentralised identity management.

To arrive at the claimed invention, it is first necessary to recognise the Applicant's problem of secure cross-chain verification of factual blockchain states; next to select features from D2 that address a different technical problem; and then to reconstruct D1 in a manner that enables credential-based verification of source-chain facts.

This reasoning proceeds with knowledge of the invention and uses the claimed architecture as a blueprint for combining the cited documents.

The fact that D2 may teach distributed storage and public-key verification does not provide any teaching, suggestion or motivation toward the core inventive concept of the present application, namely enabling a destination chain to make business decisions based upon independently verifiable assertions regarding source-chain state.

The combination therefore can only be reached with hindsight knowledge of the Applicant's disclosure.

Conclusion

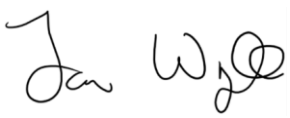
In summary, the Applicant respectfully submits that the claimed invention is not merely a combination of D1's cross-chain identity verification and D2's distributed storage architecture.

Rather, it introduces a fundamentally different verification paradigm in which verifiable credentials (VC) carry assertions regarding source-chain facts and are used to support independent verification of those facts against actual blockchain state.

Since neither D1 nor D2 teaches or suggests this concept, and because the Examiner's combination relies upon hindsight reconstruction of the Applicant's invention, the Applicant respectfully submits that Claims 1-20 involve an inventive step over the combined disclosures of D1 and D2.

Favourable reconsideration of the application is respectfully requested in light of these submissions.

Yours faithfully,
Laminar IP



Jon Wright
Registered Patent Attorney